

## DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 | 2026–2027 | Trimester – 4

### Project Abstract Submission Form

|           |             |                                                    |
|-----------|-------------|----------------------------------------------------|
| Sec No- 9 | Team No- 17 | Project Title- Online Auction and Bidding Platform |
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#### 1. Abstract

The **Online Auction & Bidding Platform** is a web-based application that provides a secure and efficient platform for buying and selling products through online auctions. Sellers can create auction listings by providing product details, starting prices, and auction durations, while buyers can view available products and place bids in real time. The system uses **React.js** for the frontend, **Node.js** and **Express.js** for the backend, and **MySQL** for storing user, auction, bidding, payment, and transaction data.

The platform uses **Socket.io** to provide real-time bidding updates so that users can see new bids instantly. An **anti-sniping** timer extends the auction when a bid is placed near the closing time, ensuring fair participation. **Redis** is used to improve the performance of frequently accessed auction data, while **Stripe** is used for secure payment processing and settlement. The system also provides separate dashboards for buyers, sellers, and administrators, along with auction verification and dispute management features.

Overall, the project aims to create a reliable, transparent, and user-friendly online auction system that manages the complete auction process, from product listing and live bidding to winner selection, payment, and settlement.

#### 2. Team Details

| Roll Number(s) | Name of the Student | Signature |
|----------------|---------------------|-----------|
| 2520031613     | U.Moksha Ratna      |           |
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#### 3. Frameworks / Platforms / Software Used

- **React.js** – Frontend development and user interface
- **Node.js** – Backend runtime environment
- **Express.js** – Backend API and server development
- **MySQL** – Database management and storage
- **Socket.io** – Real-time bidding and live auction updates
- **Redis** – Caching and real-time auction data management
- **Stripe API** – Secure online payment and settlement
- **JWT** – User authentication and authorization
- **Postman** – API testing
- **Visual Studio Code** – Development environment
- **Git & GitHub** – Version control and project management

Guide Name and Signature (remarks if any) : Dr.Prasanthi Y